Theory for a phenomenological description of the magnetoresistance of a EIO/TTO heterostructure

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We write the full model for the system as $\mathcal{H} = \mathcal{H}_{\rm EIO} + \mathcal{H}_{\rm TTO} + \mathcal{H}_{\rm int}$, where $\mathcal{H}_{\rm EIO}$ is a tight-binding Hamiltonian describing a slab of $Eu_2Ir_2O_7$, $\mathcal{H}_{\rm TTO}$ is a pseudospin-1/2 exchange Hamiltonian describing $Tb_2Ti_2O_7$ and $\mathcal{H}_{\rm int}$ captures the interaction at the interface.

1 Behavior of TTO under a magnetic field

We consider the most general nearest-neighbor symmetry-allowed exchange Hamiltonian on the pyrochlore lattice with non-Kramers doublets [1, 2]

$$\begin{aligned}\mathcal{H}_{\rm TTO} = &\sum_{\langle ij\rangle}\left\{J_{zz}\ \mathrm{S}_i^z\ \mathrm{S}_j^z - J_\pm\left(\mathrm{S}_i^+\mathrm{S}_j^- + \mathrm{S}_i^-\mathrm{S}_j^+\right) + J_{\pm\pm}\left(\gamma_{ij}\ \mathrm{S}_i^+\mathrm{S}_j^+ + \gamma_{ij}^*\ \mathrm{S}_i^-\mathrm{S}_j^-\right)\right\}\\ &- \sum_i\left[g_z h_i^z \mathrm{S}_i^z + g_\perp h_i^x \mathrm{S}_i^x + g_\perp h_i^y \mathrm{S}_i^y\right]\\ &- \delta_1\sum_i\left[h_i^x h_i^z \mathrm{S}_i^x + h_i^y h_i^z \mathrm{S}_i^y\right]\\ &- \delta_2\sum_i\left[((h_i^y)^2 - (h_i^x)^2)\mathrm{S}_i^x + 2h_i^x h_i^y \mathrm{S}_i^y\right],\end{aligned} \tag{1}$$

where S_i^α are the pseudospin components in the local $\{\hat{\mathbf{x}}_i, \hat{\mathbf{y}}_i, \hat{\mathbf{z}}_i\}$ frame and

$$\gamma = \begin{pmatrix} 0 & 1 & w & w^2\\ 1 & 0 & w^2 & w\\ w & w^2 & 0 & 1\\ w^2 & w & 1 & 0\end{pmatrix}, \tag{2}$$

with $w = e^{2\pi i/3}$. The coupling to a magnetic field $\mathbf{B} = |\mathbf{B}|\hat{\mathbf{n}}$ is written in terms of $h_i^z = \mu_B(\hat{\mathbf{z}}_i\cdot\mathbf{B})$, $h_i^x = \mu_B(\hat{\mathbf{x}}_i\cdot\mathbf{B})$ and $h_i^y = \mu_B(\hat{\mathbf{y}}_i\cdot\mathbf{B})$. The components of the g-tensor of the ions in the presence of a magnetic field $g_z \approx 10$ and $g_\perp < 1$ [3, 4]. The quadratic coupling to the transverse components is expected to be relevant for $Tb_2Ti_2O_7$, where $\delta_{1,2} \sim (1.4\ \mathrm{meV})^{-1}$[2]. We can rewrite (1) as

$$H = \sum_{\langle ij\rangle}\left\{J_{zz}\ \mathrm{S}_i^z\ \mathrm{S}_j^z - J_\pm\left(\mathrm{S}_i^+\mathrm{S}_j^- + \mathrm{S}_i^-\mathrm{S}_j^+\right) + J_{\pm\pm}\left(\gamma_{ij}\ \mathrm{S}_i^+\mathrm{S}_j^+ + \gamma_{ij}^*\ \mathrm{S}_i^-\mathrm{S}_j^-\right)\right\} - \sum_i\left[b_i^z\mathrm{S}_i^z + b_i^+\mathrm{S}_i^+ + b_i^-\mathrm{S}_i^-\right], \tag{3}$$

$$b_i^z = g_z h_i^z, \tag{4}$$

$$b_i^+ = (b_i^-)^* = \frac{g_\perp(h_i^x - ih_i^y) + \delta_1(h_i^x - ih_i^y)h_i^z + \delta_2(h_i^y - ih_i^x)^2}{2}. \tag{5}$$

Next, we calculate the dynamical spin structure factor

$$S_\mu(\mathbf{q},\omega) = \frac{1}{2\pi N_{\text{u.c.}}}\langle \mathbf{S}_{\mu,\mathbf{q},\omega}\cdot\mathbf{S}_{\mu,-\mathbf{q},-\omega}\rangle \tag{6}$$

$$= \frac{1}{2}S^{+-}_{\mu\mu}(\mathbf{q},\omega) + \frac{1}{2}S^{-+}_{\mu\mu}(\mathbf{q},\omega) + S^{zz}_{\mu\mu}(\mathbf{q},\omega), \tag{7}$$

where $\mu = 0,1,2,3$ labels the pyrochlore sublattice and

$$S^{ab}_{\mu\nu}(\mathbf{q},\omega) = \frac{1}{2\pi N_{\text{u.c.}}}\int_{-\infty}^{\infty} dt\; e^{i\omega t}\left\langle S^a_\mu(\mathbf{q},t)S^b_\nu(-\mathbf{q},0)\right\rangle. \tag{8}$$

As the magnetic field increases, the system enters a polarized state in which the spins coupled to the field are aligned along the local z direction. Previous works have used spin wave theory to estimate the values of the exchange couplings in $Yb_2Ti_2O_7$ by fitting to inelastic neutron scattering experimental data [1, 5, 6]. Here we proceed in a similar fashion in order to calculate $S_\mu(\mathbf{q},\omega)$.

First, we define new "quantization" frames in which the z direction is the orientation of the spins that classically minimize the Hamiltonian (1), $\hat{\mathbf{z}}^*_\mu = 2\;\mathbf{S}_{\text{cl},\mu}$. Then we choose $\hat{\mathbf{x}}^*_\mu = \hat{\mathbf{z}}^*_\mu \times (1,1,1)/|\hat{\mathbf{z}}^*_\mu \times (1,1,1)|$ and $\hat{\mathbf{y}}^*_\mu = \hat{\mathbf{z}}^*_\mu \times \hat{\mathbf{x}}^*_\mu$. The transformation from the local frames to the quantization frames corresponds to rotation matrices with elements R^{ab}_μ (for instance, $R^{33}_\mu = \hat{\mathbf{z}}^*_\mu\cdot\hat{\mathbf{z}}_\mu$ and $R^{12}_\mu = \hat{\mathbf{x}}^*_\mu\cdot\hat{\mathbf{y}}_\mu$). Then, we expand the Hamiltonian around the ordered state by expressing the spin operators in terms of Holstein-Primakoff (HP) bosons

$$\tilde{S}^z_i = s - n_i, \tag{9}$$

$$\tilde{S}^+_i = \sqrt{2s-n_i}\,b_i \approx \sqrt{2s}\;b_i, \tag{10}$$

$$\tilde{S}^-_i = b^\dagger_i\sqrt{2s-n_i} \approx \sqrt{2s}\;b^\dagger_i, \tag{11}$$

where $n_i = b^\dagger_i b_i$ and we have retained only the leading terms in a $1/s$ expansion to capture the quadratic fluctuations. To improve notation, we rewrite the Hamiltonian as $\mathcal{H} = \sum_{\langle ij\rangle}\mathcal{J}^{ab}_{ij}S^a_i S^b_j$, where summation over repeated indices is implied. Transforming to the quantization frame yields

$$\mathcal{H} = \sum_{\mathbf{r},\mathbf{r}'}\sum_{\mu,\nu}\mathcal{J}^{ab}_{\mathbf{r}-\mathbf{r}',\mu\nu}R^{ca}_\mu R^{db}_\nu \tilde{S}^c_{\mathbf{r}+\mathbf{e}_\mu}\tilde{S}^d_{\mathbf{r}'+\mathbf{e}_\nu}. \tag{12}$$

After substituting the HP bosons, the Hamiltonian reads as

$$\mathcal{H} \approx \mathcal{H}_0 + s\mathcal{H}_2 + s^{1/2}\mathcal{H}_3 + s^0\mathcal{H}_4, \tag{13}$$

where $\mathcal{H}_0 = N_{\text{u.c.}}s(s+1)E_{\text{cl}}$ is the classical energy contribution and $\mathcal{H}_n$ corresponds to the n-magnon interactions. Now we introduce the bosonic operator $\boldsymbol{b}_\mathbf{k} = \left(b_{\mathbf{k}0}, b_{\mathbf{k}1}, b_{\mathbf{k}2}, b_{\mathbf{k}3}, b^\dagger_{-\mathbf{k}0}, b^\dagger_{-\mathbf{k}1}, b^\dagger_{-\mathbf{k}2}, b^\dagger_{-\mathbf{k}3}\right)$, where the Fourier components are defined by the transformation $b_{\mathbf{R}_\mu} = \frac{1}{\sqrt{N_{\text{u.c.}}}}\sum_\mathbf{k} b_{\mathbf{k}\mu}\;e^{i\mathbf{k}\cdot\mathbf{R}_\mu}$. With the Fourier transformed exchange couplings defined as $\mathcal{J}^{ab}_{\mathbf{k},\mu\nu} = \sum_{\boldsymbol{\delta}}\mathcal{J}^{ab}_{\boldsymbol{\delta},\mu\nu}e^{-i\mathbf{k}\cdot(\boldsymbol{\delta}+\mathbf{e}_\mu/2-\mathbf{e}_\nu/2)}$, the quadratic Hamiltonian takes the form

$$\mathcal{H}_2 = \sum_\mathbf{k}\boldsymbol{b}^\dagger_\mathbf{k}H_\mathbf{k}\boldsymbol{b}_\mathbf{k}, \tag{14}$$

where

$$H_\mathbf{k} = \begin{pmatrix}\boldsymbol{A}_\mathbf{k} & \boldsymbol{B}_\mathbf{k}\\ \boldsymbol{B}^*_{-\mathbf{k}} & \boldsymbol{A}^*_{-\mathbf{k}}\end{pmatrix}, \tag{15}$$

with the matrix elements given by

$$A^{\mu\nu}_\mathbf{k} = 2\mathcal{J}^{-+}_{\mathbf{k},\mu\nu} - \delta_{\mu\nu}\sum_\sigma\mathcal{J}^{zz}_{\mathbf{0},\mu\sigma} + \delta_{\mu\nu}\sum_{a=z,\pm}b^a_\mu R^{3a}_\mu, \tag{16}$$

$$B^{\mu\nu}_{\mathbf{k}} = 2\mathcal{J}^{--}_{\mathbf{k},\mu\nu}. \tag{17}$$

Here, $R^{\pm z}_{\mu} = (R^{xz}_{\mu} \mp iR^{yz}_{\mu})/2$. When diagonalizing the Hamiltonian (15), we must be careful to preserve the commutation relations for the bosonic operators, which can be written as $[\boldsymbol{b}^{\alpha}_{\mathbf{k}}, \boldsymbol{b}^{\dagger\beta}_{\mathbf{q}}] = \delta_{\mathbf{k},\mathbf{q}}\eta^{\alpha\beta}$ and $[\boldsymbol{b}^{\alpha}_{\mathbf{k}}, \boldsymbol{b}^{\beta}_{\mathbf{q}}] = \delta_{\mathbf{k},-\mathbf{q}}\zeta^{\alpha\beta}$, where

$$\eta = \begin{pmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{0} & -\mathbf{1} \end{pmatrix}, \quad \zeta = \begin{pmatrix} \mathbf{0} & \mathbf{1} \\ -\mathbf{1} & \mathbf{0} \end{pmatrix}. \tag{18}$$

This can be achieved by taking the four positive eigenvalues $\omega_{m,\mathbf{k}}$ and correspondent eigenvectors $v_{m,\mathbf{k}}$ of $2\eta H_{\mathbf{k}}$, imposing the normalization $v^{\dagger}_{m,\mathbf{k}}\eta v_{m',\mathbf{k}} = \eta_{mm'}$.

Since Eq. (6) is invariant under local rotations, Eq. (7) retains its form. We can rewrite Eq. (8) as

$$S^{\alpha\beta}_{\mu\nu}(\mathbf{q},\omega) = \frac{1}{N_{\text{u.c.}}}\sum_{e}\langle 0|S^{\alpha}_{\mu}(\mathbf{q})|e\rangle\langle e|S^{\beta}_{\nu}(-\mathbf{q})|0\rangle\delta(\omega-\omega_e), \tag{19}$$

where $|e\rangle$ are all the possible excited states with energies ω_e. Summing over all one-magnon states, we obtain

$$S^{+-}_{\mu\mu}(\mathbf{q},\omega) = S^{-+}_{\mu\mu}(\mathbf{q},\omega) = 2s\sum_{m} v^{\mu}_{m,\mathbf{q}}v^{\mu *}_{m,\mathbf{q}}\delta(\omega-\omega_{m,\mathbf{q}}), \tag{20}$$

and similarly summing over the ground state and all two-magnon states we obtain

$$S^{zz}_{\mu\mu}(\mathbf{q},\omega) = N_{\text{u.c.}}s^2\delta_{\mathbf{q}} + \frac{1}{N_{\text{u.c.}}}\sum_{m,\mathbf{k}}\sum_{m',\mathbf{k}'} v^{\mu}_{m,\mathbf{k}}v^{\mu *}_{m,\mathbf{k}}v^{\mu}_{m',\mathbf{k}'}v^{\mu *}_{m',\mathbf{k}'}\ \delta_{\mathbf{k}+\mathbf{k}'+\mathbf{q}}\ \delta(\omega-\omega_{m,\mathbf{k}}-\omega_{m',\mathbf{k}'}), \tag{21}$$

where the first term corresponds to the expected Bragg peak and the second term corresponds to the continuous contribution from magnon pair creation/annihilation processes.

2 Surface states in EIO

3 Magnetoresistance response

We model the interfacial interaction via a Kondo coupling between the spins located at the Tb and Ir kagome lattices

$$\mathcal{H}_{\text{int}} = \tag{22}$$

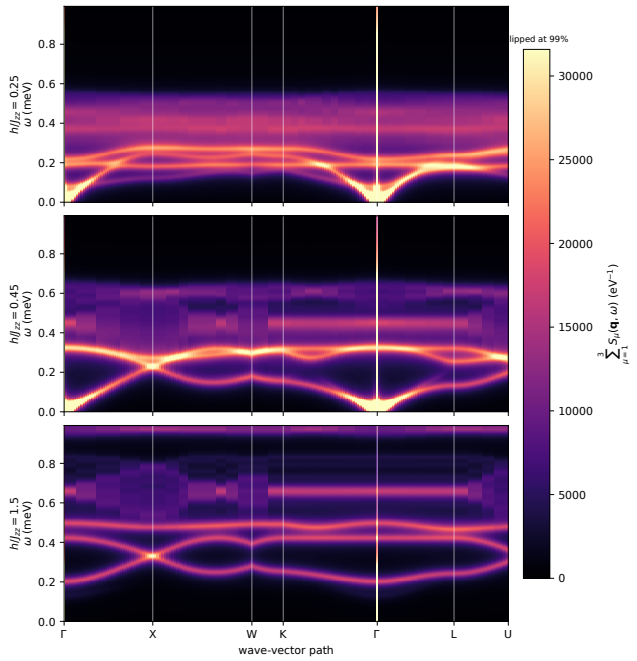
The relaxation time of mode n and momentum $\mathbf{k}$ in the case where surface states are scattered by dynamical spins is given by

$$\tau^{-1}(n,\mathbf{k}) = \sum_{ss'}\int \mathrm{d}\mathbf{k}' \int d\omega\delta\left(\epsilon_{n\mathbf{k}'} - \epsilon_{n\mathbf{k}} - \omega\right)\left|T_{\mathbf{k}\mathbf{k}',ss'}(\omega)\right|^2(1-\cos\theta_{\mathbf{k}\mathbf{k}'}) \tag{23}$$

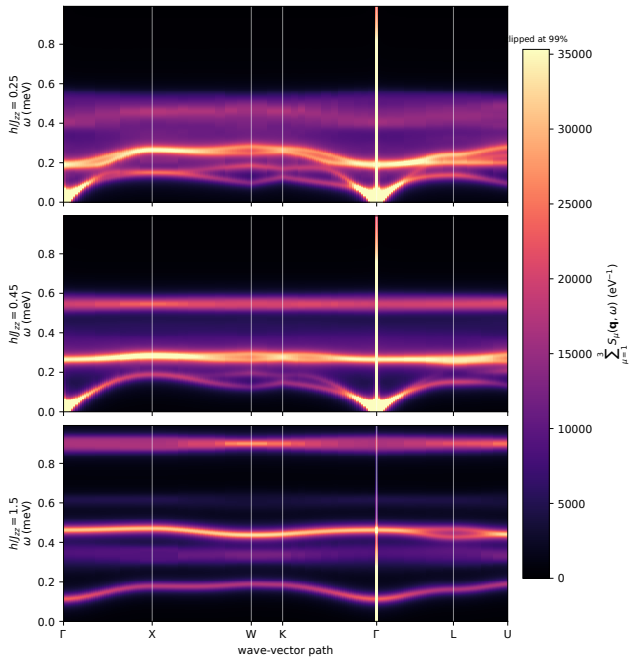
$$= \int \mathrm{d}\mathbf{k}'\left|T_{\mathbf{k}\mathbf{k}'}(\epsilon_{n\mathbf{k}'} - \epsilon_{n\mathbf{k}})\right|^2(1-\cos\theta_{\mathbf{k}\mathbf{k}'}), \tag{24}$$

where

$$|T_{\mathbf{k}\mathbf{k}'}(\omega)|^2 = |J|^2\sum_{\langle i,j\rangle} S_j(\mathbf{k}-\mathbf{k}',\omega)\sum_{s}\left|\langle i\alpha s|\mathbf{k}\rangle\langle\mathbf{k}'|i\alpha s\rangle\right|^2. \tag{25}$$



(a) $\phi = 0°$



(b) $\phi = 30°$

Figure 1: Spin structure factor $S(\mathbf{q}, \omega) = \sum_{\mu=1,2,3} S_\mu(\mathbf{q}, \omega)$ as defined in Eq (6) for $\mathbf{q}$ following a path along the high symmetry points in the BZ. The exchange couplings have been taken as $J_{zz} = 0.2\text{meV}, J_\pm = 0.2J_{zz}$ and $J_{\pm\pm} = 0$.

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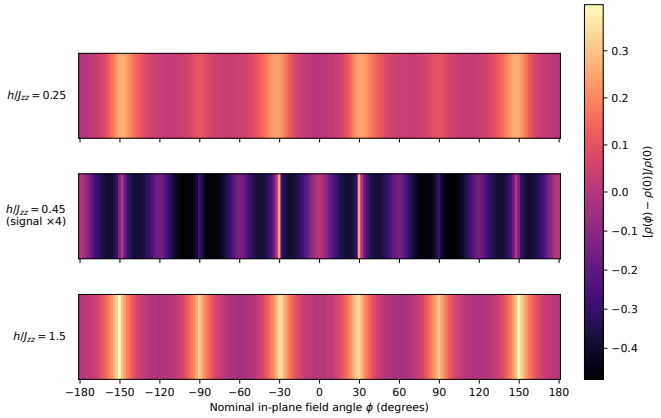
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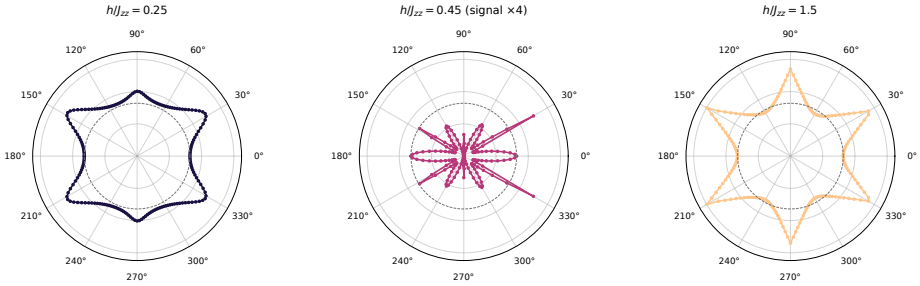
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(a)



(b)

Figure 2: Magnetoresistance $\mathrm{MR}_\phi = [\rho(\phi) - \rho(0)]/\rho(0)$ evaluated at $h/J_{zz} = 0.25, 0.45$ and 1.5. The exchange couplings have been taken as $J_{zz} = 0.2\mathrm{meV}, J_\pm = 0.2J_{zz}$ and $J_{\pm\pm} = 0$.